

Vital Signs Simulator Build and Test

NAME _____ DATE _____ PERIOD _____

The boundary

This program is a **training simulation**. It does not measure a person, diagnose a condition, or replace medical equipment. Use only fictional values supplied by the lesson.

Build in Microsoft MakeCode

1. Open **makecode.microbit.org** and create a project named Vital Signs Simulator .
2. In on button A pressed ,set heartRate to pick random 60 to 110 .
3. Add show number heartRate .
4. Add a heart LED image, pause 200 ms , and a smaller or blank heart image.
5. In on button B pressed ,set temperature to pick random 970 to 1005 .
6. Add show number temperature and explain that 986 represents 98.6°F because the micro:bit display does not show decimals in this build.
7. Test both buttons in the on-screen simulator.
8. Save a screenshot or share link. Download to a physical micro:bit only if one is available.

Test record

TEST	EXPECTED RESULT	WHAT HAPPENED	REVISION MADE
Button A	Shows a fictional heart-rate value		
Button A visual	Shows a pulse animation		
Button B	Shows a fictional temperature code		
Repeat test	Values can change within the assigned range		

Add a handoff alert

Add this **simulation rule** after Button A displays the value:

- If heartRate > 100 , show the letter R for **report**.
- Otherwise, show a check mark.

The R does not diagnose anything. It reminds the learner to document the value, listen to the fictional patient, and report through the scenario's chain of supervision.

ALERT TEST	FICTIONAL INPUT	EXPECTED OUTPUT	PASS?
Within assigned range	84	Check mark	
Report branch	108	R	

Explain your design

Which block stores a value? _____

Which block makes a decision? _____

Why is this program a simulator rather than a medical device?

How does accurate testing connect to nursing work?

Equal evidence routes

- ☐ Physical micro:bit
- ☐ MakeCode browser simulator
- ☐ Paper block trace supplied by the teacher

The route does not change the score. Evidence is the tested logic and explanation.